



Wrocław
University
of Science
and Technology

Car Rental Management System

Course: Software System Development, Project, group 1

Instructor: dr inż. Marcin Jodłowiec

Team members:

Wojciech Wojnarowski 263594

Aleksandra Król 272530

Leon Wyrodek 293852

Youssef Ayadi 294635

1. Project description

1.1 Project business goals

The Car Rental Management System is a software product developed with the intent to be sold to car rental companies. The development team acts as a software vendor, not as a car rental business operator. The product is designed to be purchased and deployed by client companies, who will use it to manage their own rental operations.

The main objective of the Car Rental Management System is to deliver a software product that automates and streamlines the operations of a car rental company, improving efficiency, customer experience, and business control.

Automate rental operations

- A customer should be able to complete a full reservation in under 5 minutes
- Payment processing and confirmation should require no manual intervention from staff
- Rental agreements should be generated automatically upon payment confirmation

Enhance customer experience

- Customers should be able to filter and sort vehicles by type, make, model, drive type, and price
- Customers should be able to view detailed vehicle information before making a reservation
- The platform should be accessible via any modern web browser without additional software installation

Enable business reporting

- The system should allow the generation of reports on rental activity for any selected time period
- Key business metrics such as fleet utilisation and revenue should be visible in the reporting module

1.2 Identification of the project's internal and external stakeholders

Internal stakeholders

Symbol	Name	Role	Description
PM	Leon Wyrobek Smeder	Project Manager	Responsible for planning, organizing, and coordinating the project activities
BE	Wojciech Wojnarowski	Backend Developer	Responsible for server-side logic and application functionality
FE	Aleksandra Król	Frontend Developer	Responsible for developing the user interface and ensuring a good user experience
DB	Youssef Ayadi	Database Specialist	Responsible for designing and managing the project database

External stakeholder

Symbol	Name	Role	Description
CUST	End Users	Customer	People who rent cars using the system expect easy booking and reliability
PAY	Payment Provider	Payment Service	Handles online payments, transactions, and refunds
INS	Insurance Company	Insurance Provider	Provides insurance coverage and

			manages claims
GOV	Government Authorities	Regulatory Body	Ensures compliance with legal regulations and data protection laws
PART	Business Partners	Fleet Provider	Supplies vehicles and may integrate with the system
SERV	Maintenance Services	Service Provider	Responsible for vehicle maintenance and repair updates
API	External API Providers	Technology Provider	Provides services like maps, GPS tracking, or identity verification

1.3 Domain description

The car rental domain involves providing vehicles to customers for temporary use in exchange for payment. The Car Rental Management System supports and automates the main processes of a car rental company by handling customer interactions, reservations, payments, and operational workflows.

The system behavior can be described through a sequence of key events:

Vehicle Search and Selection

- A customer **accesses the system** via a web interface.
- The customer **searches for available vehicles** based on criteria such as rental dates, vehicle type, price, or model.
- The system **retrieves and displays a list of available vehicles** matching the search criteria.
- The customer **views detailed information** about selected vehicles, including specifications and pricing.

Reservation Creation and Payment

- The customer **selects a vehicle** and **defines the rental period**.

- The system **calculates the total rental cost** and presents the rental conditions.
- The customer **enters personal information** and **submits a reservation request**.
- The system **initiates a payment transaction** through an external payment provider.
- Upon successful payment:
 - The system **confirms the reservation**.
 - The system **generates a rental agreement automatically**.
 - The system **updates vehicle availability**.

Vehicle Pickup Process

- At the start of the rental period, an employee **retrieves the reservation details**.
- The employee **verifies the customer's identity and reservation**.
- The employee **inspects the vehicle condition** before handover.
- The system **records the pickup event** and **updates the rental status to active**.

Rental Monitoring and Support

- During the rental period, the system may **track the vehicle location** using GPS integration.
- The system **maintains insurance coverage information** associated with the rental.
- External systems (e.g., insurance providers) may **handle incidents or claims** if needed.

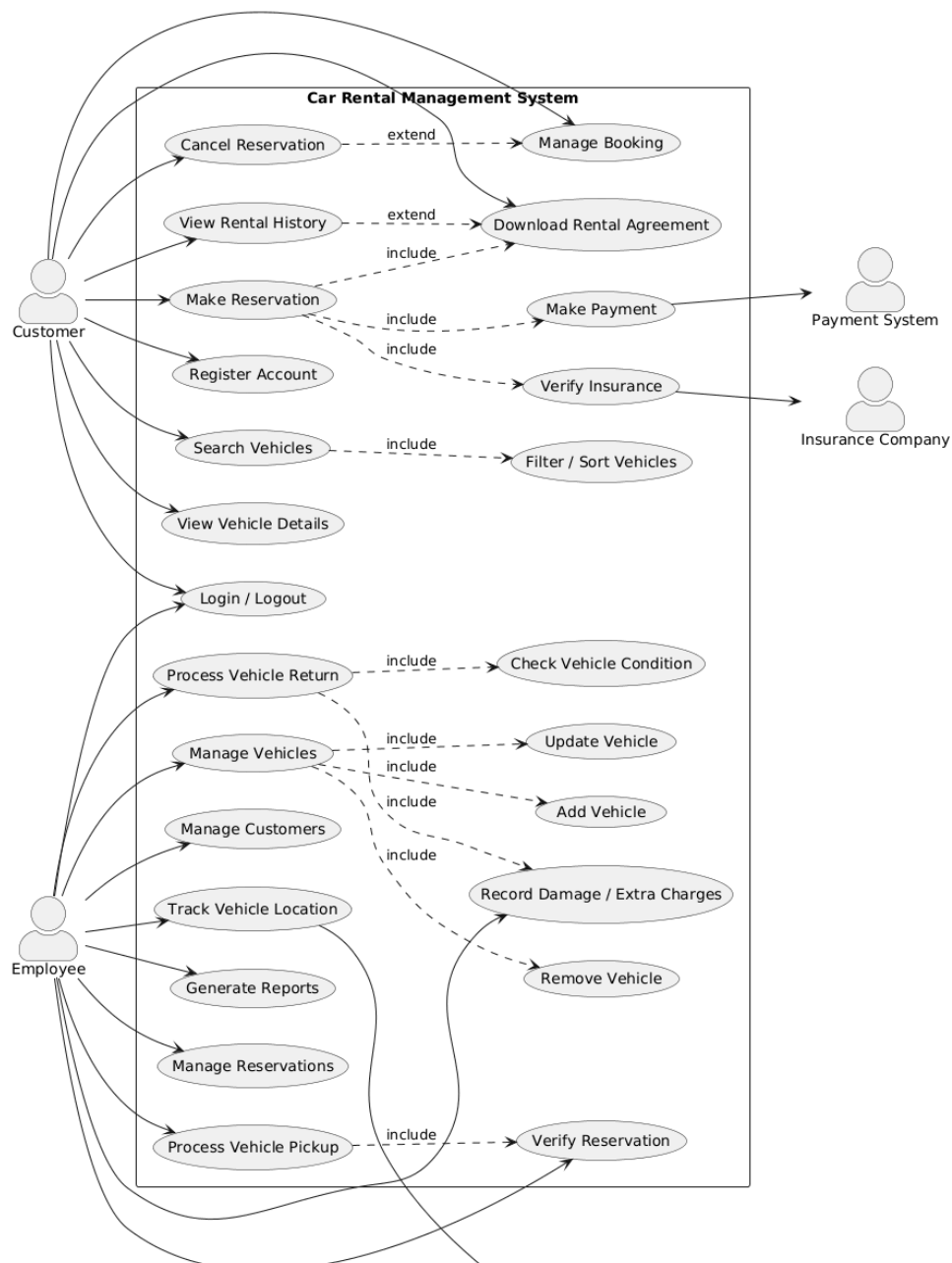
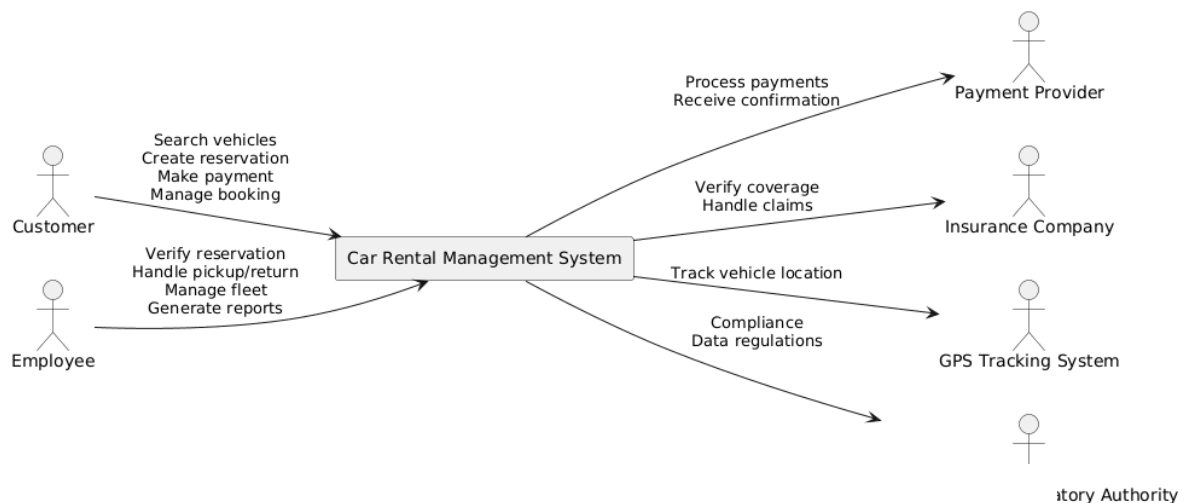
Vehicle Return and Rental Completion

- At the end of the rental period, the customer **returns the vehicle**.
- The employee **inspects the vehicle's condition** after return.
- The system **records any damages or additional charges**, if applicable.
- The system **finalizes the rental transaction** and **updates vehicle availability**.

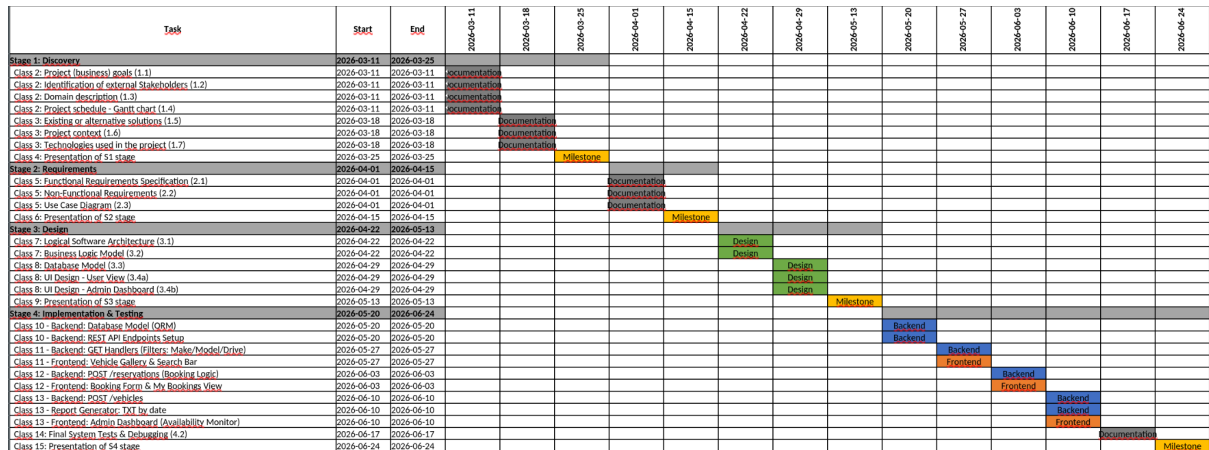
Administrative and Management Activities

- Employees **manage customer data, vehicle records, and reservations**.
- Employees can **add, update, or remove vehicles** from the fleet.
- The system **generates reports and analytics** on rental activity, revenue, and fleet utilization.

Overall, the system coordinates interactions between customers, employees, and external services such as payment providers, insurance companies, and GPS systems. By structuring the domain around these events, the system ensures a clear, efficient, and automated rental process.



1.4 Gantt Chart



1.5 Identification of existing or alternative solutions

We have identified 3 different competitors to our application, with each of them having unique characteristics and features:

1. Kaizen Rent (<https://kaizenrent.pl>)
2. Car Net (<https://carnet.pl>)
3. Sixt (<https://www.sixt.pl>)

Kaizen Rent

- **Name of the project/application:** Kaizen Rent
- **Main features:**
 - Fast online reservation system with real-time availability.
 - Wide range of car classes from economy to premium and SUVs.
 - Add-on management (GPS, child seats, additional drivers).
 - Local branch-specific search (e.g., Wroclaw Airport vs. City Center).
- **Advantages:**
 - **User-friendly interface:** The booking process is streamlined and requires very few steps to complete.
 - **Local focus:** Highly optimized for the Polish market.
 - **Clear pricing:** Transparent breakdown of costs before reaching checkout.
- **Disadvantages:**

- **Limited global presence:** Primarily focused on Poland, which may not appeal to international travelers.
- **Standard Fleet:** Lacks the highly specialized "luxury" or "niche" vehicles found in global premium competitors.
- **Lacks drive type as a filtering option**
- **Shows minimal information about the car**
- **You can get a similar car to the model you choose;** however, unlike Sixt, they specify which cars count as similar

CarNet

- **Name of the project/application:** CarNet
- **Main features:**
 - Short-term, medium-term, and long-term rental options.
 - Van and specialized vehicle rental (not just passenger cars).
 - "Delivery to customer" service feature during the booking process.
 - Transparent deposit information and "No-deposit" options for specific segments.
- **Advantages:**
 - **Fleet Versatility:** Excellent selection of transport vehicles and vans, great for both individuals and businesses.
 - **Competitive Pricing:** Often offers better rates for long-term domestic rentals compared to international brands.
 - **Flexible Pick-up:** A large number of local points across Polish cities.
- **Disadvantages:**
 - **Website Design:** The interface feels slightly off and less "modern" than Sixt or Kaizen Rent.
 - **App Integration:** The mobile experience is less polished than that of other competitors.
 - **Reservation by calling:** No option to reserve a car with the web app
 - **Minimal filtering options**
 - **Shows minimal information about the car**
 - **You can get a similar car to the model you choose**

Sixt

Name of the project/application: SIXT

Main features:

- Integrated mobility platform: car rental, car sharing, and ride-hailing in one app.
- Premium fleet focus (BMW, Audi, Mercedes-Benz).
- Sixth Fastlane: Skip-the-counter feature using digital keys on a smartphone.
- Global loyalty program with status-based upgrades.

Advantages:

- **Modern Technology:** One of the best mobile apps in the industry with seamless digital check-in.
- **Premium Brand Image:** High-quality, newer vehicles with high specifications.
- **Global Reach:** A single account works in almost every country in the world.
- **Shows the price upfront**
- **Nice minimalistic interface:** easy to look at for the user

Disadvantages:

- **Cost:** Generally, one of the most expensive options on the market.
- **Strict Terms:** High security deposits on credit cards and complex insurance/extra-fee structures that can significantly increase the final price.
- **Minimal filtering options**
- **Shows minimal information about the car**
- **You can get a similar car to the model you choose**

1.6 Project context

Application context

The Car Rental Management System is designed to support and automate the daily operations of a car rental company. The system serves two primary user groups: customers, who interact with the platform to browse vehicles, make reservations, and manage their bookings online; and employees, who use the system to manage the fleet, process rentals, verify reservations, and generate reports. The system addresses common inefficiencies found in small and medium-sized rental businesses, such as manual reservation handling, a lack of real-time vehicle availability tracking, and limited digital interaction with customers. The platform is intended to be accessible via a web browser, requiring no dedicated hardware installation on the customer side.

Technological context

The system is developed as a standalone software product intended to be purchased and deployed by a car rental company. The hosting environment is therefore the responsibility of the client organisation. The system architecture should be designed to be deployment-agnostic, supporting common hosting setups

including cloud platforms (such as AWS, Azure, or Google Cloud) and on-premise servers. Clear deployment documentation will be provided to ensure the client can set up and maintain the system independently. The system should be designed to be scalable, allowing it to handle increasing numbers of users and reservations. The architecture should support deployment in cloud environments with the possibility of horizontal scaling.

Organizational context

The system is being developed by a four-person software development team acting as a vendor, with defined roles: Project Manager, Backend Developer, Frontend Developer, and Database Specialist. The product is intended to be sold to car rental companies, who will act as the operating organisation once the system is deployed

Legal context

As the system operates within the Polish and broader EU market, it is subject to several legal frameworks. The General Data Protection Regulation (GDPR) applies directly, as the system collects and stores personal data, including customer names, contact details, and payment information. This requires that data be processed lawfully, stored securely, and that customers be informed of how their data is used. The right to access, correct, and delete personal data must be supported. Online payment processing falls under the EU Payment Services Directive (PSD2), which mandates Strong Customer Authentication (SCA) for electronic transactions. Additionally, rental agreements constitute legally binding contracts, meaning the system must produce accurate and complete documentation. If driver's license verification is implemented, relevant identification data handling regulations must also be observed.

1.7 Technologies used in the project

Java: A widely used, object-oriented programming language serving as the primary language for the backend of the system.

- **Key responsibilities:** Backend business logic and application layer
- **Link:** <https://www.java.com>

Spring Boot: An open-source Java-based framework that simplifies the development of production-ready web applications and REST APIs.

- **Key responsibilities:** REST API development, application configuration, and dependency management
- **Link:** <https://spring.io/projects/spring-boot>

Hibernate: A Java-based ORM framework that handles communication between the application and the relational database, mapping Java objects to database tables automatically.

- **Key responsibilities:** Database access layer, entity mapping, and schema generation
- **Link:** <https://hibernate.org>

MySQL: An open-source relational database management system that stores all persistent data, including customers, vehicles, reservations, and payments.

- **Key responsibilities:** Persistent data storage and management
- **Link:** <https://www.mysql.com>

React: An open-source TypeScript library for building dynamic and responsive user interfaces using a component-based architecture.

- **Key responsibilities:** Frontend user interface, customer booking flow, and employee dashboard
- **Link:** <https://react.dev>